

The Internet & Its Uses

Syllabus

5.1 Internet & World Wide Web (WWW)

- 1 Understand the difference between the internet and the World Wide Web
 - The internet is the infrastructure
 - The world wide web is the collection of websites and web pages accessed using the internet
- 2 Understand what is meant by a uniform resource locator (URL)
 - A URL is a text-based address for a web page; it can contain the protocol, the domain name and the web page/file name
- 3 Describe the purpose and operation of hypertext transfer protocol (HTTP) and hypertext transfer protocol secure (HTTPS)
- 4 Explain the purpose and functions of a web browser
 - The main purpose of a web browser is to render hypertext markup language (HTML) and display web pages
 - Functions include:
 - Storing bookmarks and favorites
 - Recording user history
 - Allowing use of multiple tabs
 - Storing cookies
 - Providing navigation tools
 - Providing an address bar
- 5 Describe how web pages are located, retrieved and displayed on a device when a user enters a URL
 - Including the role of:
 - Web Browser
 - IP Addresses
 - Domain Name Server (DNS)
 - Web Server
 - HTML
- 6 Explain what is meant by cookies and how they are used, including session cookies and persistent cookies
 - Cookies are used for functions, including:
 - Saving personal details
 - Tracking user preferences
 - Holding items in an online shopping cart
 - Storing login details



5.2 Digital Currency

- 1 Understand the concept of a digital currency and how digital currencies are used
 - 2 Understand the process of blockchain and how it is used to track digital currency transactions
- A digital currency is one that only exists electronically
 - Blockchain, in its basic form, is a digital ledger, that is a time-stamped series of records that cannot be altered

5.3 Cyber Security

- 1 Describe the processes involved in, and the aim of carrying out, a range of cyber security threats.
 - Including:
 - Brute-Force Attack
 - Data Interception
 - Distributed Denial of Service (DDoS) Attack
 - Hacking
 - Malware (virus, worm, Trojan horse, spyware, adware, ransomware)
 - Pharming
 - Phishing
 - Social Engineering
 - ➡ Including:
 - Access Levels
 - Anti-malware including Anti-virus and Anti-spyware
 - Authentication (username and password, biometrics, two-step verification)
 - Automating software updates
 - Checking the spelling and tone of communications
 - Checking the URL attached to a link
 - Firewalls
 - Privacy settings
 - Proxy-servers
 - Secure Socket Layer (SSL) security protocol
- 2 Explain how a range of solutions are used to help keep data safe from security threats.



5.1 The Internet & The World Wide Web

Internet: Derived from two words **INTER**connected **NET**work (**INTERNET**), collection of Interconnected Networks. Relies on physical infrastructure that allows networks and devices to connect to other networks and devices.

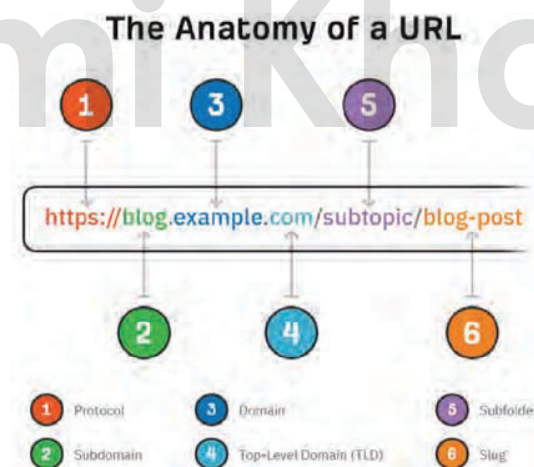
World Wide Web (WWW): Part of the Internet that users can access using Web Browser Software, is a collection of web-pages, which is based on the Hypertext Transfer Protocol. It is a way of accessing information using Internet.

Internet Vs. World Wide Web

Internet	World Wide Web
• Send & Receive Emails • Allow online chatting • Makes use of Transmission Control Protocols (TCP) & Internet Protocols (IP) • Worldwide collection of interconnected networks and devices	• Collection of multimedia web pages and other information on websites • HTTP(s) protocols are written using Hypertext Mark-up Language (HTML) • Uniform Resource Locator (URLs) are used to specify the location of web pages • Web resources are accessed by web browsers • Uses the internet to access information from web servers

Uniform Resource Locator (URL)

Web Browsers are usually just referred to as browsers. Software that allow users to access and display web pages on their device screens. Browsers interpret hypertext mark-up language (HTML) sent from websites and produce the results on the user's device. Uniform resource locators (URLs) are text addresses used to access websites. A URL is typed into a browser address bar using the following format:



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HTTP and HTTPS

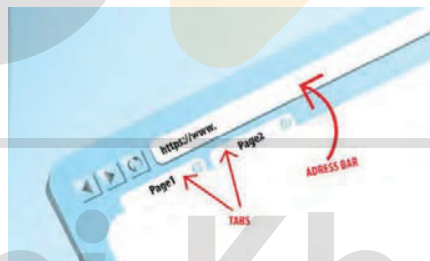
Hypertext Transfer Protocol (HTTP) is a set of rules that must be obeyed when transferring files across the internet. When some form of security (for example, SSL or TLS) is used, then this changes to https. The 's' stands for secure, and indicates a more secure way of sending and receiving data across a network.

Web Browsers

Browsers are software that allow a user to access and display web pages on their device screens. Browsers interpret (translate) the HTML from websites and show the result of the translation.

Features

- Home Page
- Favorite Website/Webpages (Bookmarks)
- User History
- Navigation
- Multiple Tabs
- Cookies
- Hyperlinks
- Cache
- Use of Javascript
- Address Bar



Retrieval & Location of Web Pages

HTML (HyperText Markup Language) is a language used to display content on browsers. All websites are written in HTML and hosted on a web server that has its own IP address. To retrieve pages from a website your browser needs to know this IP address. The Domain Name Server (DNS) (also known as domain name system) is a system for finding IP addresses for a domain name given in a URL. URLs and domain name servers eliminate the need for a user to memorize IP addresses. The DNS process involves converting a URL (such as www.hoddereducation.co.uk) into an IP address the computer can understand (such as 107.162.140.19). The DNS process involves more than one server.

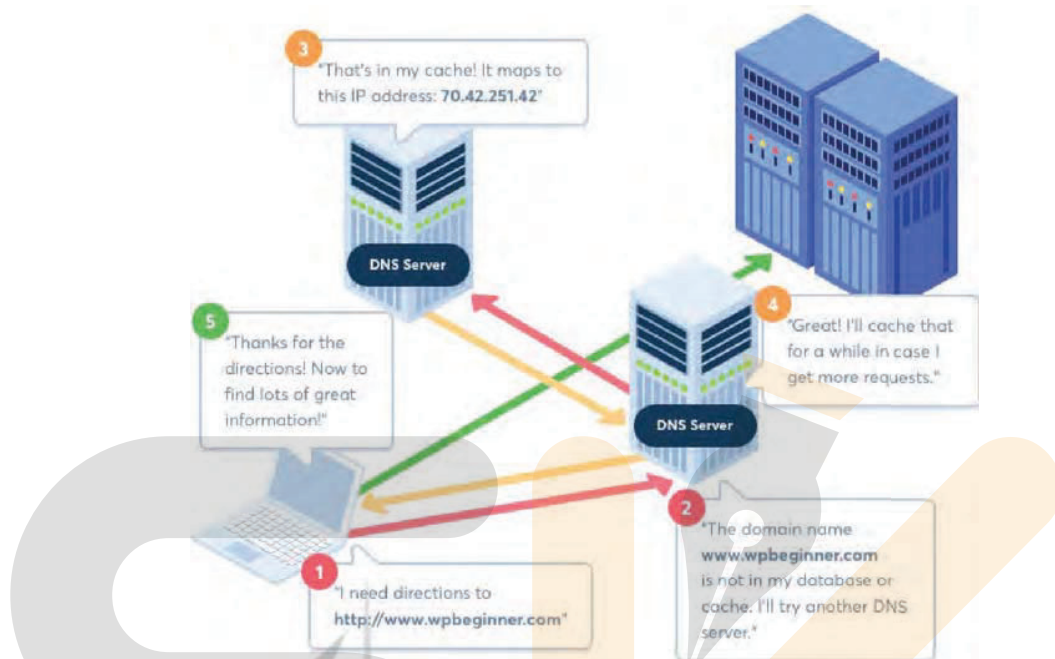
DNS servers contain a database of URLs with the matching IP addresses.



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Cookies

Cookies are small files or code stored on a user's computer. They are sent by a web server to a browser on a user's computer. Each cookie is effectively a small look-up table containing pairs of (key, data) values, for example, (surname, Jones) (music, rock). Every time a user visits a website, it checks if it has set cookies on their browser before.

There are two types of cookies:

- ➔ Session Cookie
- ➔ Persistent (Permanent) Cookie

Session Cookies

Session cookies are used, for example, when making online purchases. They keep a user's items in a virtual shopping basket. This type of cookie is stored in temporary memory on the computer, doesn't actually collect any information from the user's computer and doesn't personally identify a user.

Persistent Cookies

Persistent cookies remember a user's log in details (so that they can authenticate the user's browser). They are stored on the hard drive of a user's computer until the expiry date is reached or the user deletes it. These cookies remain in operation on the user's computer even after the browser is closed or the website session is terminated.



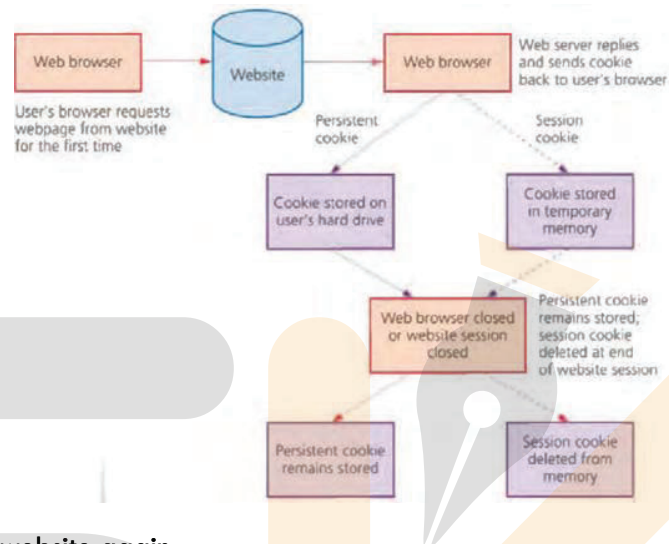
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Following summarizes subsequent visits:

- ➔ First time the user logs in to website



- ➔ User logs in to website again

Uses of Persistent Cookies

- ➔ Remember authentication
- ➔ Serve as a memory
- ➔ Save users' items in a virtual shopping basket/cart
- ➔ Track internet habits
- ➔ Target users with advertising
- ➔ Store users' preferences
- ➔ Online Financial Transactions
- ➔ Progress in online games and quizzes to be stored
- ➔ Social networking sites to recognize certain preferences and browsing histories
- ➔ Different languages



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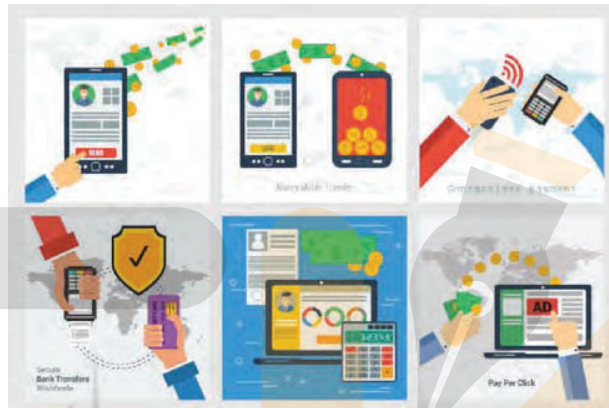


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5.2 Digital Currency

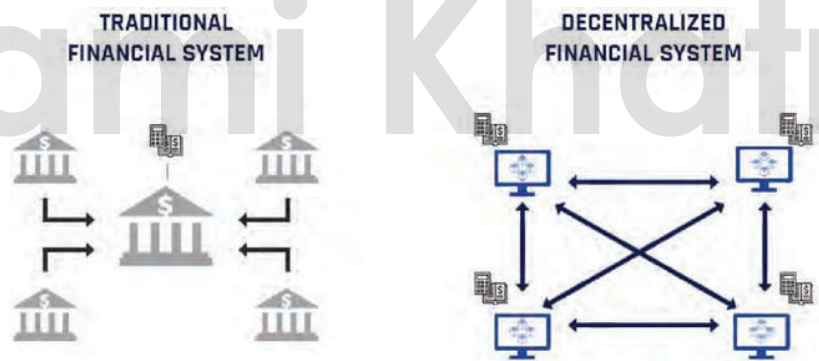
Digital Currency

Digital money, or digital currency, is any form of money or payment that exists only in electronic form.



Digital money can be transferred and exchanged with technologies like credit cards, smartphones, and online cryptocurrency exchanges.

Digital Currency is dependent on a central banking system BUT the biggest concern in this is of security and confidentiality. This concern is addressed and solution for this is Decentralization, Cryptocurrency.



- ➔ Cryptocurrency uses cryptography to track transaction.
- ➔ Cryptocurrency has no state control and all the rules are set by the cryptocurrency community itself.



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- Cryptocurrency transactions are publicly available and therefore all transactions can be tracked and the amount of money in the system is monitored.
- Cryptocurrency system works by being within a blockchain network which means it is much more secure.

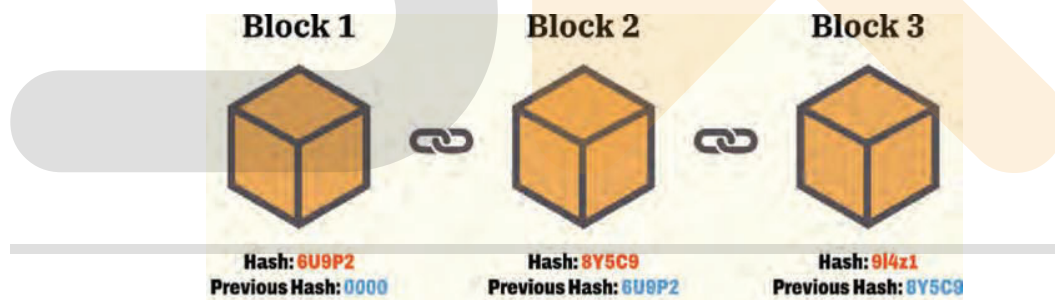
Blockchain

Blockchain is a decentralized, distributed and public digital ledger that is used to record transactions across many computers so that the record cannot be altered retroactively without the alteration of all subsequent blocks and the consensus of the network. Uses of Blockchain are:

- Cryptocurrency exchanges
- Smart contracts
- Research (particularly within pharmaceutical companies)
- Politics
- Education

How Blockchain Works?

Each block is created. This hash value is unique to each block and includes a timestamp, which identifies when an event actually takes place. We will now consider what happens when a chain of blocks is created



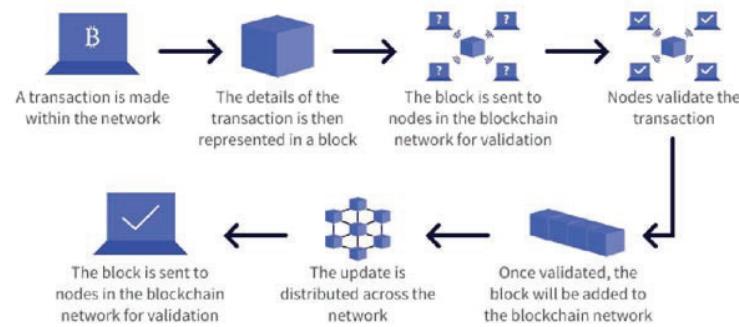
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First Block in a Blockchain in a Blockchain is known as the 'Genesis Block' which is not pointing to any previous block. In Blockchain every next block is connected to previous block through previous hash.

This will prevent tampering. However, it may be crossing your mind that computers are now so fast that it should be possible to quickly create a whole new string of blocks, and therefore recreate a new chain before the problem has been discovered. Prevented by Proof of Work (POW). POW makes sure it takes ten minutes to determine the necessary proof-of-work for each block before it can be added to the chain.

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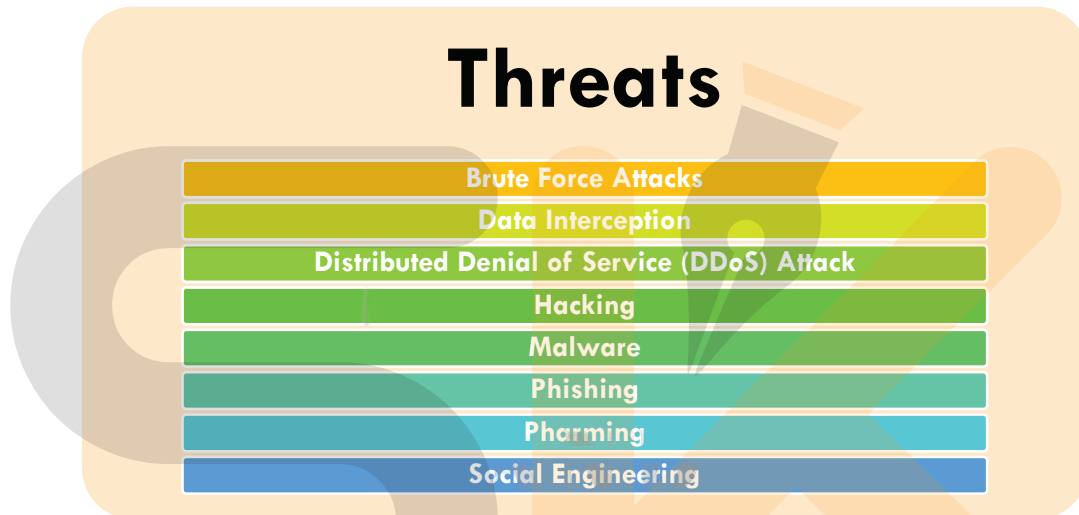
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5.3 Cyber Security

Cyber Security Threats

Keeping your data safe is an important concern in modern world.

Data can be corrupted or deleted either through accidental damage or malicious acts. There are also many ways data can be intercepted leading to cyber security threats.



Brute Force Attacks

Brute Force Attack is a hacking technique that uses trial and error to crack passwords, login credentials, and encryption keys. It is a simple yet reliable tactic for gaining unauthorized access to individual accounts and organizations' systems and networks. The hacker tries multiple usernames and passwords, often using a computer to test a wide range of combinations, until they find the correct login information.

One way to reduce the number of attempts needed to crack a password is to first go through a series of Logical methods:

Method 1: Check if the password is one of the most common ones used (the five most common are: name, 1234567, password, qwerty, 1111111 and abcd123).

Method 2: Start with a strong word list (this is a text file containing a collection of words that can be used in a brute force attack); some programs will generate a word list containing a million words.

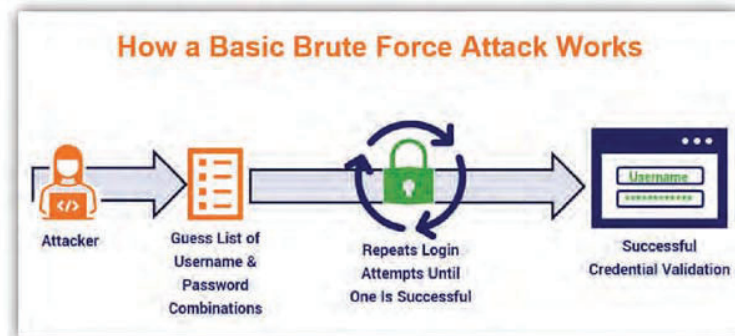
Method 2 is time consuming method as it will take several hours. The longer the password and character variation used, harder it is to create



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Data Interception

Data Interception is a form of stealing data by tapping into a wired or wireless communication link. The intent is to compromise privacy or to obtain confidential information.

Interception can be carried out using a packet sniffer, which examines data packets being sent over a network. The intercepted data is sent back to the hacker. This is a common method when wired networks are used.

Distributed Denial of Service (DDoS) Attack

Denial-of-Service (DoS) Attack is an attack meant to shut down a machine or network, making it inaccessible to its intended users. The attackers may be able to prevent a user from:

- Accessing their emails
- Accessing websites/webpages
- Accessing online services

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In a Distributed Denial of Service (DDoS) the spam traffic originates from many different computers, which makes it hard to block the attack.

This can happen to a user's email account, for example, by an attacker sending out many spam messages to their email account. Internet service providers (ISPs) only allow a specific data quota for each user. Consequently, if the attacker sends out thousands of emails to the user's account, it will quickly become clogged up and the user won't be able to receive legitimate emails. An individual user or a website can guard against these attacks to some degree by:

- ➔ Using an up-to-date malware checker
- ➔ Setting up a firewall to restrict traffic to and from the web server or user's computer
- ➔ Applying email filters to filter out unwanted traffic (for example, spam)

There are certain signs a user can look out for to see if they have become a victim of a DDoS attack:

- ➔ Slow network performance (opening files or accessing certain websites)
- ➔ Inability to access certain websites
- ➔ Large amounts of spam email reaching the user's email account

Hackin

Hacking is the act of identifying and then exploiting weaknesses in a computer system or network, usually to gain unauthorized access to personal or organizational data. Hacking is not always a malicious activity, but the term has mostly negative connotations due to its association with cybercrime.

Malicious hacking, takes place without the user's permission, and is always an illegal act. However, universities and companies now run courses in ethical hacking. This occurs when companies authorize paid hackers to check out their security measures and test how robust their computer systems are to hacking attacks.

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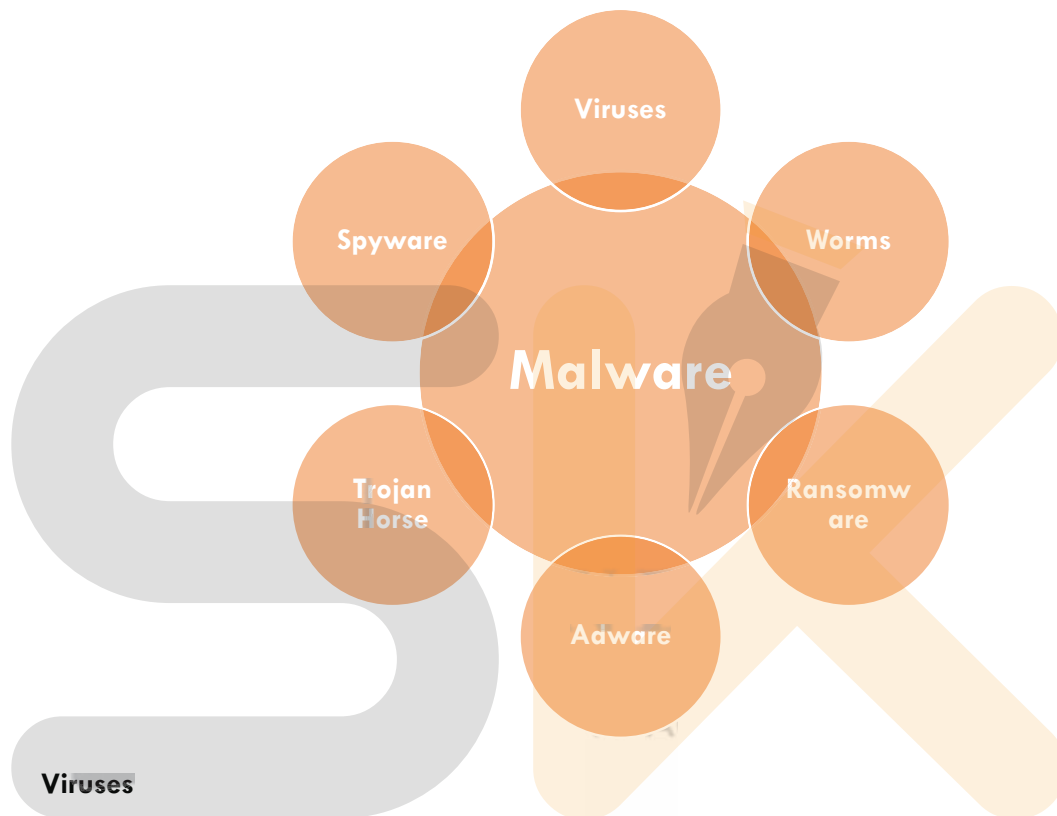
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Malware

Malware is one of the biggest risks to the integrity and security of data on a computer system. There are many types of malware:



Viruses

Viruses are programs or program code that replicate (multiply) with the intention of deleting or corrupting files, or causing a computer to malfunction.

Viruses need an active host program on the target computer or an operating system that has already been infected, before they can actually run and cause harm.



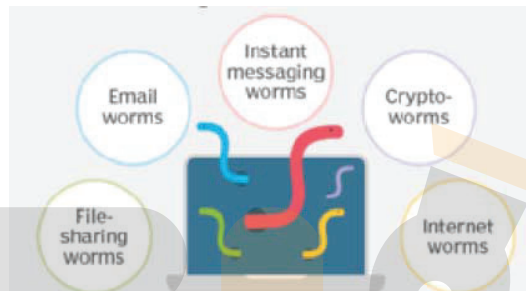
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Worms

Worm Virus refers to a malicious program that replicates itself, automatically spreading through a network. In this definition of computer worms, the worm virus exploits vulnerabilities in your security software to steal sensitive information, install backdoors that can be used to access the system, corrupt files, and do other kinds of harm.

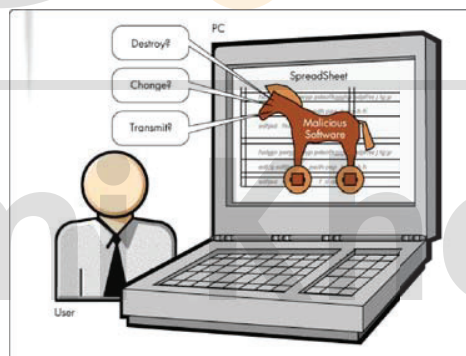


Trojan Horse

Trojan Horse is a program which is often disguised as legitimate software but with malicious instructions embedded within it. A Trojan horse replaces all or part of the legitimate software with the intent of carrying out some harm to the user's computer system.

They need to be executed by the end-user and therefore usually arrive as an email attachment or are downloaded from an infected website.

Once installed on the user's computer, the Trojan horse will give cyber criminals access to personal information on your computers, such as IP addresses, passwords and other personal data.



Spyware

Program installed on a PC to gather information about the user. It monitors every key press and relays the information back to the home base.

Adware

Adware is a type of malware. At its least dangerous it will attempt to flood an end-user with unwanted advertising. For example, it could redirect a user's browser to a website that



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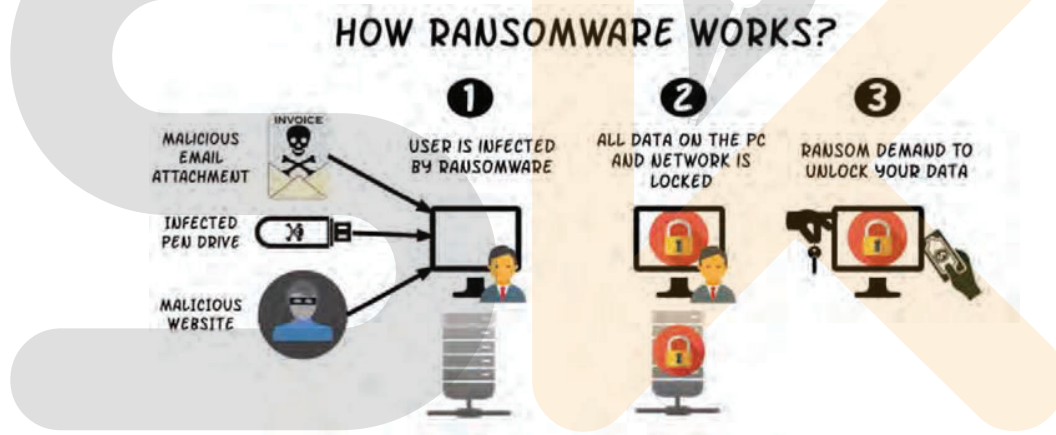
contains promotional advertising, it could appear in the form of pop-ups, or it could appear in the browser's toolbar and redirect search requests.

Adware can:

- Highlight weaknesses in a user's security defenses
- Hard to remove
- Hijack a browser

Ransomware

Ransomware are programs that encrypt data on a user's computer and hold the data hostage. The cybercriminal waits until the ransom money is paid and, sometimes, the decryption key is then sent to the user. It has caused considerable damage to some companies and individuals.



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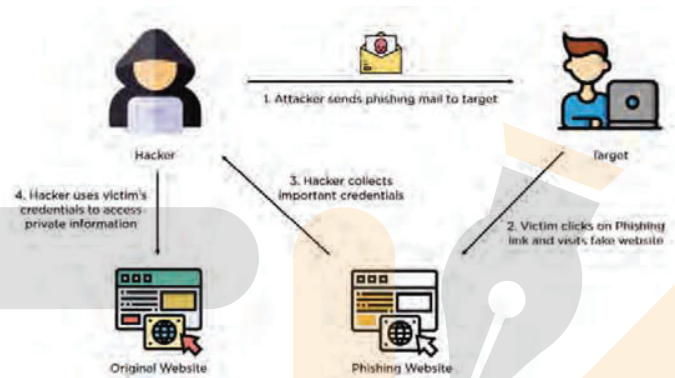
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Phishing

Phishing is a cybercrime in which a target or targets are contacted by email, telephone or text message by someone posing as a legitimate institution to lure individuals into providing sensitive data such as personally identifiable information, banking and credit card details, and passwords.



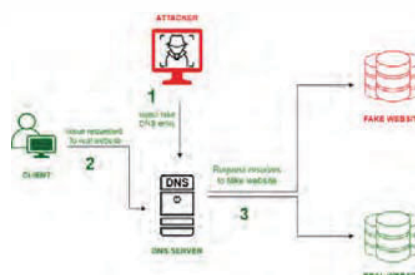
The key point is that the recipient has to initiate some act before the phishing scam can cause any harm.

Ways to help prevent Phishing Attacks:

- Users need to be aware of new phishing scams.
- Not to click on any email links unless totally certain that it is safe to do so.
- Important to run anti-phishing toolbars on browse.
- Important to run anti-phishing toolbars on browse.
- Regular checks of online accounts.
- Ensure up-to-date browser is running on the computer device.
- Be very wary of pop-ups and use the browser to block them.

Pharming

Pharming is malicious code installed on a user's computer or on an infected website. The code redirects the user's browser to a fake website without the user's knowledge. Unlike phishing, the user doesn't actually need to take any action for it to be initiated. The creator of the malicious code can gain personal data, such as bank details, from the user. Often the website appears to come from a trusted source and can lead to fraud and identity theft.



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Why does pharming pose a threat to data security?

Pharming redirects internet users to a fake or malicious website set up by, for example, a hacker; redirection from a legitimate website to the fake website can be done using DNS cache poisoning.

Social Engineering

Social engineering occurs when a cybercriminal creates a social situation that can lead to a potential victim dropping their guard. It involves the manipulation of people into breaking their normal security procedures and not following best practice. There are five types of threat that commonly exist:

Instant Messaging

- Malicious links are embedded into instant messages; for example, an important software upgrade (relies on user's curiosity).

Scareware

- Using a Pop-up message that claims that the user's computer is infected with a virus; the user is told that they need to download the fake anti-virus immediately (relies on user's fear).

Email/Phishing Scams

- User is tricked by the apparent genuineness of an email and opens a link in the email; this redirects web browser to a fake website (relies on user's trust of well-known companies).

Baiting

- Cybercriminal leaves a malware infected memory stick somewhere where it can be found; the finder picks up the memory stick and plugs it into their computer and unwittingly downloads malicious software (relies on user's curiosity).

Phone Calls

- For example, so called IT Professional calls the user on their mobile claiming their device has been compromised in some way; the user is advised to download some special software that allows the cybercriminals to take over the user's device giving them access to personal information (relies on fear).

The whole idea is based on the exploitation of certain human emotions; the three most common ones to exploit are:

- Fear
- Curiosity
- Empathy & Trust

Access Levels

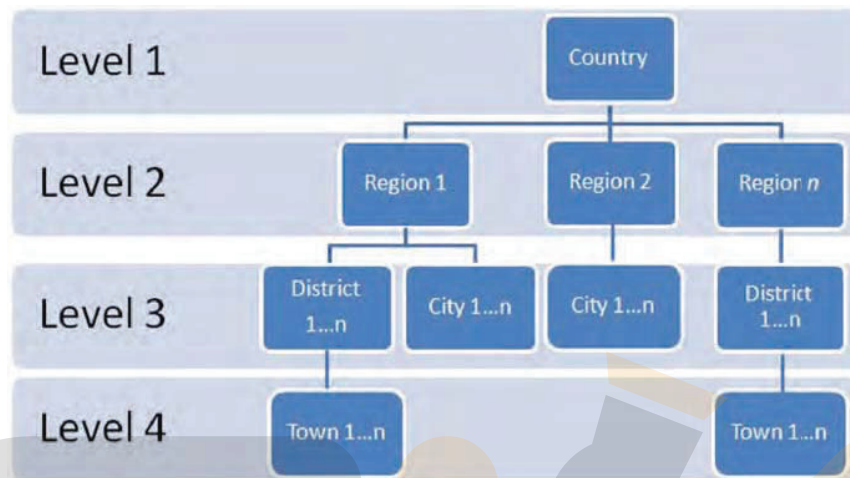
Every user is given with different Access Levels & Rights like in organization different departments have different access levels an HR Person is not given access to Accounts files.



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Another area where access levels are very important is in social network. Usually, four access levels i.e., Public Access, Friends, Custom, Data owner.

Antimalware

Two most common types are: Antivirus & Antispyware.

Antispyware

Anti-spyware software detects and removes spyware programs installed illegally on a user's computer system. The software is based on one of the methods: Rules, File Structures.

General Features are:

- Detect and remove spyware already installed on a device.
- Prevent a user from downloading spyware.
- Encrypt files to make the data more secure in case it is 'spied' on.
- Encryption of keyboard strokes to help remove the risk posed by the keylogging aspects of some spyware.
- Blocks access to a user's webcam and microphone.
- Scans for signs that the user's personal information has been stolen and warns the user if this has happened.

Authentication

Authentication refers to login credentials. There are three common factors used in authentication: something you know, something you have, something which is unique to you.

Passwords & Usernames

Passwords and Usernames (Login Credentials) are used to restrict access to data or systems. They should be hard to crack and changed frequently to retain any real level of security. Passwords can also take the form of biometrics.



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Biometrics

Biometrics are rising as an advanced layer to many personal and enterprise security systems. With the unique identifiers of your biology and behaviors, this may seem foolproof. However, biometric identity has made many cautious about its use as standalone authentication.

Fingerprint Scans

A fingerprint scanner is a type of electronic security system that uses fingerprints for biometric authentication to grant a user access to information or to approve transactions.

Images of fingerprints are compared against previously scanned fingerprint images stored in a database; if they match, then a user has been correctly recognized. The system compares patterns of 'ridges' and 'valleys' that are unique. The accuracy of the scan is about around 1 in 5000. Fingerprint scanning techniques have the following benefits as a form of security:

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- Unique
- Other security devices can be lost or even lost which makes them less effective.
- Can't be misplaced
- Since it is unique nobody else can authenticate using fingerprints

**Drawbacks**

- Expensive
- If finger is damaged/injured, it can effect the accuracy
- Infringement of civil liberties

**Advantages**

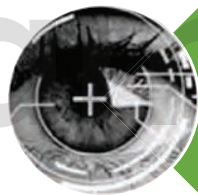
- Small storage requirements
- Easy to use
- Most developed biometric techniques

Retina Scans

Retina scanning is a biometric verification technology that uses an image of an individual's retinal blood vessel pattern as a unique identifying trait for access to secure installations. It is very secure since nobody has yet found a way to duplicate the blood vessels patterns. The accuracy is about 1 in 10 million.

**Advantages**

- High accuracy
- Replication not possible

**Drawbacks**

- Intrusive
- Slow
- Expensive

Face Recognition

Facial recognition is a way of identifying or confirming an individual's identity using their face. Facial recognition systems can be used to identify people in photos, videos, or in real-time.



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Drawbacks

- Affected by changes in lighting, the person's hair, change in age, and if the person is wearing glasses.



Advantages

- Non-Intrusive
- Inexpensive

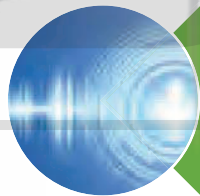
Voice Recognition

Voice recognition is a computer software program or hardware device with the ability to decode the human voice. Voice recognition is commonly used to operate a device, perform commands, or write without having to use a keyboard, mouse, or press any buttons.



Drawbacks

- Low accuracy
- Used for unauthorized access
- Illness such as cold can change a person's voice

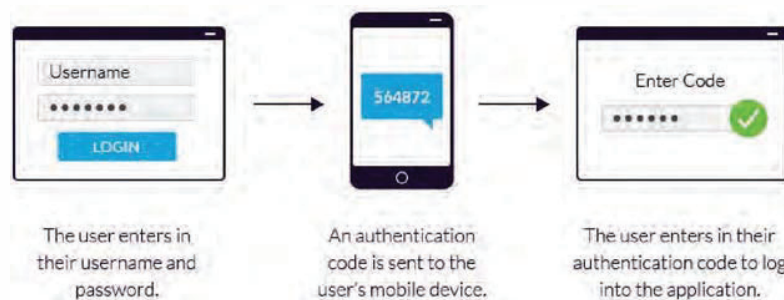


Advantages

- Non-Intrusive
- Less time consuming
- Inexpensive

Two Step Verification

Two-step verification requires two methods of authentication to verify who a user is. It is used predominantly when a user makes an online purchase using a credit/debit card as payment method.



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Automatic Software Updates

Automatic software updates mean software on computers and mobile phones/ tablets is kept up-to-date. Sometimes this is done overnight or when you log off the device.

These updates are vital since they may contain patches that update the software security or improve the software performance.

Checking the Spelling & Tone of Communication and URL Links

When emails are sent to you, there are three actions you always need to take before opening them or activating any links in them.

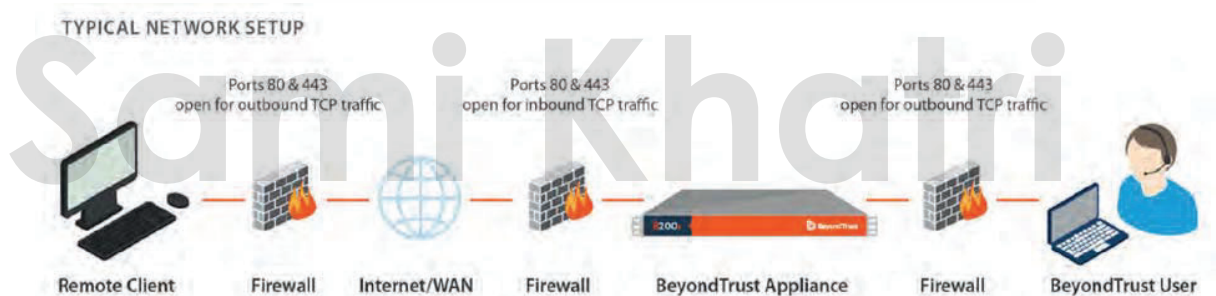
- Check out the spellings in the email and in the links.
- Carefully check the tone in the email message.

There are five things to look out for:

1. The email address itself; no legitimate company will use an email address such as: @gmail.com.
e.g., account-update@gmail.com
2. The tone of the email and bad spelling of words is a clear indication of a potential scam.
3. Misspelling of domain names in a link are very common errors found in emails sent by scammers and fraudsters.
4. Suspicious links
5. Other errors to look out for are just plain spelling mistakes.

Firewalls

A firewall can be either software or hardware. It sits between the user's computer and an external network and filters information in and out of the computer.



Main tasks:

- To examine the traffic between user's computer and a public network.
- Check whether incoming or outgoing data meets a given set of criteria.



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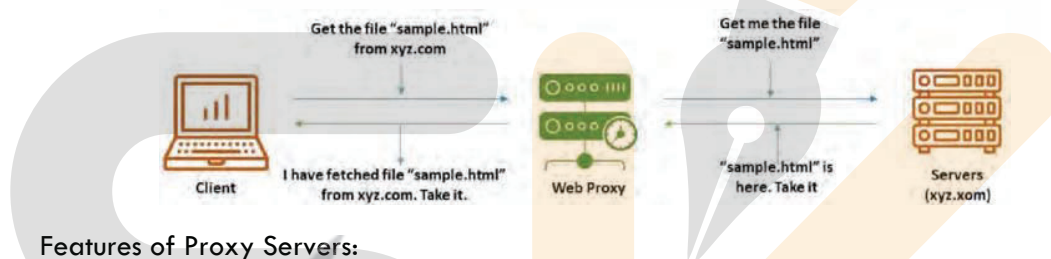


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- Firewall will block the traffic and give the user a warning that there may be a security issue.
- Firewall can be used to log all incoming and outgoing traffic to allow later interrogation by the user.
- Criteria can be setup so the firewall prevents access to certain undesirable sites.
- Firewalls to help prevent viruses or hackers entering the user's computer.
- User is warned if some software on their system trying to access an external data source.

Proxy's Server

Proxy servers act as an intermediate between the user and a web server:



Features of Proxy Servers:

- Allows internet traffic to be filtered
- Keeps users' IP addresses secret which improves security
- If the internet traffic is valid, access to the web server is allowed
- If the internet traffic is invalid, access to the web server is denied
- It is possible to block requests from certain IP addresses
- Prevents direct access to a web server by sitting between the user and the web server
- If an attack is launched, it hits the proxy server instead
- Used to direct invalid traffic away from web servers which gives additional protection
- Using the feature known as a cache, it is possible to speed up access to information/data from a website.

- Proxy servers can also act as firewalls

Privacy Settings

Privacy settings are the controls available on web browsers, social networks and other websites that are designed to limit who can access and see a user's personal profile.

Privacy settings can refer to:

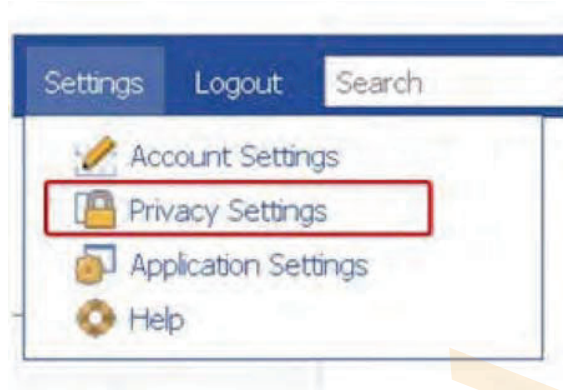
- Do not track settings
- Check to see if payment methods have been saved on websites
- Safer browsing
- Web browser privacy options
- Website advertising opt-outs
- Apps



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Secure Sockets Layers (SSL)

Secure Sockets Layer (SSL) is a type of protocol - a set of rules used by computers to communicate with each other across a network. This allows data to be sent and received securely over the internet.

SSL is a form of digital certificate which means any communication or data exchange between browser and website is secure provided this certificate is provided.

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Applications

- Online Banking & Online Financial Transactions
- Online Shopping
- Sending software out to a restricted list of users
- Sending and receiving emails
- Cloud storage facilities
- Intranet and Extranet
- VOIP
- Instant Messaging
- Social Networking Site

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